the grind never stops

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September 20, 2024

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1 Induction and Pigeonhole

Problem 1.1

Prove for all positive integers n the identity

$$\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{2n}=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}-\cdots+\frac{1}{2n-1}-\frac{1}{2n}.$$

Solution. We induct on n. The base case $n=1$ is obvious, as

$$\frac{1}{1+1}=\frac{1}{1}-\frac{1}{2}=\frac{1}{2}.$$

Suppose now that

$$\frac{1}{n+1}+\frac{1}{n+2}+\cdots+\frac{1}{2n}=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}-\cdots+\frac{1}{2n-1}-\frac{1}{2n}.$$

Then, add $\frac{1}{2n+1}+\frac{1}{2n+2}-\frac{1}{n+1}$ to both sides, getting the equality

$$\begin{aligned}\frac{1}{n+2}+\cdots+\frac{1}{2n+1}+\frac{1}{2n+2}&=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}-\cdots-\frac{1}{2n}+\frac{1}{2n+1}+\left(\frac{1}{2n+2}-\frac{1}{n+1}\right)\\&=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}-\cdots-\frac{1}{2n}+\frac{1}{2n+1}+\frac{1}{2n+2}-\frac{2}{2n+2}\\&=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}-\cdots-\frac{1}{2n}+\frac{1}{2n+1}-\frac{1}{2n+2}\end{aligned}$$

which is exactly the desired equality with $n+1$ in place of n, so we are done by the principle of mathematical induction. □

Problem 1.2

Let n be a positive integer. Show that

$$1+\frac{1}{2^3}+\frac{1}{3^3}+\cdots+\frac{1}{n^3}<\frac{3}{2}.$$

All of the following solutions rely on the idea that the left-hand side can be bounded above by the obviously convergent series

$$1+\frac{1}{2^3}+\frac{1}{3^3}+\cdots=\sum_{k=1}^{\infty}\frac{1}{n^3}.$$

Solution 1. Since $\frac{1}{x^3}$ is decreasing, the integral

$$1+\int_1^{\infty}\frac{1}{x^3}\,dx=\frac{3}{2}$$

is an upper bound for the series because the series is just the right Riemann sum for the integral. □

Solution 2. Let S be the sum of all the terms with even denominator, so that the series we are trying to bound is $8S$. Then, by bounding $\frac{1}{(2n)^3} > \frac{1}{(2n+1)^3}$, we can say that the desired series is bounded above by

$$8S < 1 + 2S.$$

This tells us $S < \frac{1}{6}$, so $8S < \frac{4}{3} < \frac{3}{2}$. □

Solution 3. Similar to the previous solution, bound the series by

$$1 + \frac{2}{2^3} + \frac{4}{4^3} + \frac{8}{8^3} + \cdots,$$

where each $\frac{1}{k^3}$ is overestimated by $\frac{1}{(2^m)^3}$ where 2^m is the largest power of 2 that is at most k. This is then a geometric series summing to $\frac{4}{3} < \frac{3}{2}$, so we have the necessary upper bound. □

Solution 4. For $k > 1$, bound

$$\frac{1}{k^3} < \frac{1}{k^3 - k} = \frac{1}{2(k-1)} + \frac{1}{2(k+1)} - \frac{1}{k}.$$

The sum

$$1 + \sum_{k=2}^{\infty} \frac{1}{k^3 - k},$$

which overestimates the desired series, thus telescopes to $\frac{5}{4} < \frac{3}{2}$, so we are done. □

Solution 5. Bound above by

$$1 + \frac{1}{2^3} + \frac{1}{3^3} + \sum_{k=4}^{\infty} \frac{1}{k^2}.$$

It is well known that the last addend is equal to $\frac{\pi^2}{6} - 1 - \frac{1}{4} - \frac{1}{9}$, and we can manually check (e.g. by bounding $\pi^2 < 10$) that this upper bound is less than $\frac{3}{2}$. □

Problem 1.3

Let $x_1, x_2, \ldots, x_n, y_1, y_2, \ldots, y_m$ be positive integers, with $m, n > 1$. Assume that $x_1 + \cdots + x_n = y_1 + \cdots + y_m < mn$. Prove that in the expression

$$x_1 + \cdots + x_n = y_1 + \cdots + y_m$$

you can suppress some but not all of the terms and still have equality.

Solution. Suppose that $x_1 \le x_2 \le \cdots \le x_n$ and $y_1 \le y_2 \le \cdots \le y_m$, and that $x_i \ne y_j$ for all i, j (otherwise we are done by deleting an equal pair of terms), and let $S = \sum x_i = \sum y_i$.

We induct on m and n simulatneously, showing that the if the statement is true when $(m, n) = (a-1, b)$ and $(a, b-1)$ then it is also true when $(m, n) = (a, b)$.

First, as the base case, we show that the statement holds whenever $m = 2$ (symmetry says that the same logic works for $n = 2$). To do so, we induct on n, with the base case $n = 2$ being trivial. We consider two cases:

- Suppose that some x_a satisfies $2 \le x_a \le x_2$. Then, delete x_a and replace y_2 with $y_2 - x_a$. Clearly the new set of x_is has the same sum as the new y_is, which is strictly bounded above by $2n - 2 = 2(n-1)$, and we have reduced the number of x_is, so induction finishes.

- Otherwise, every x_i is either larger than x_2 or smaller than 1. Since $x_2 \geq \frac{S}{2}$, only x_n can be larger than y_2; all the others must be equal to 1, and indeed this forces $x_n > y_2 \geq \frac{S}{2}$. However, this situation is then impossible because $\sum x_i = x_n + n - 1$ and $y_2 \geq \lceil \frac{x_n+n-1}{2} \rceil$, but the right-hand side is always at least as large as x_n because $x_n \leq n$.

Now, the rest of the problem is simple: either $x_n < y_m$, in which case we delete x_n and replace y_m with $y_m - x_n$ to reduce to $(m, n-1)$, or we reduce to $(m, n-1)$ in the same way. ☐

Problem 1.4

no.

Problem 1.5

The vertices of a convex polygon are colored by at least three colors so that no two consecutive vertices have the same color. Prove that you can dissect the polygon into triangles by diagonals that do not cross and whose endpoints have different colors.

Solution. Very simple induction, actually — we only need to be able to always chop a triangle off. Supposing the polygon has n sides, the base case $n = 3$ is trivial.

Then, consider an n-gon $P_1 P_2 \cdots P_n$. If there exists a color that is only assigned to one vertex P_a, then just draw the edge $P_a P_{a+2}$ and look at the n-gon obtained by deleting P_{a+1}, which is still convex and still has at least three color; namely, P_{a-1} and P_{a-2} both differ in color from each other and from P_a, since P_a was assumed to be the only vertex of its color. The inductive hypothesis then applies, and we are done.

Otherwise, consider the pairs (P_i, P_{i+2}) with indices mod n. Suppose FTSOC each pair is monochromatic; then, if n is odd, all the vertices are the same color, or, if n is even, then there are at most two colors in use, both of which violate the assumptions about the polygon. Thus, there exists a pair (P_a, P_{a+2}) with differing colors; draw this edge and delete P_{a+1}. By assumption, P_{a+1} is not the only vertex of its color, so we have not decreased the number of available colors, and thus the inductive hypothesis applies here, too. ☐

Problem 1.6

A sequence of m positive integers contains exactly n distinct terms. Prove that if $2^n \leq m$ then there exists a block of consecutive terms whose product is a perfect square.

Solution. Let $x_1, \ldots, x_n$ be the possible n distinct values in the sequence. Then, replace the kth term in the sequence with the tuple $(0, 0, \ldots, 0, 1, 0, \ldots, 0)$ whose ith entry is 1 if and only if the the term in the sequence is x_i. Treat these tuples as vectors in $\mathbb{F}_2^n$ and let v_1, $v_2, \ldots, v_m$ be the sequence of vectors.

Now, consider the sequence of partial sums v_1, $v_1 + v_2$, $v_1 + v_2 + v_3, \ldots, v_1 + \cdots + v_m$. If this sequence has two equal values, then we are done: specifically, if $i < j$,

$$v_1 + \cdots + v_i = v_1 + \cdots + v_j \implies v_{i+1} + \cdots + v_j = (0, 0, \ldots, 0),$$

meaning that the product of the terms in the sequence corresponding to the v_k between v_{i+1} and v_j is a square. More explicitly, the product is the product of $x_i^{e_i}$ over all i, where e_i is the number of times that 1 appears in the ith entry within our range of terms — since our vectors are in $\mathbb{F}_2^n$, the sum being the zero vector means that the e_i are all even, hence square. Similarly, if it contains the zero vector, we are also done. There are only $2^n - 1$ nonzero vectors, so pigeonhole tells us that two of the $m \geq 2^n$ vectors are equal and nonzero if no zero vectors are in the partial sums, which is what we wanted. □

Problem 1.7

Let p be a prime number and a, b, c be in $\mathbb{F}_p$ such that a and b are nonzero. Prove that there exist $x, y \in \mathbb{F}_p$ such that $ax^2 + by^2 = c$.

Solution. If $c = 0$ then $x = y = 0$ works. Suppose $c \neq 0$.

If c isn't a QR mod p, multiply the equation by some (nonzero) non-QR to get the equivalent equation

$$a'x^2 + b'y^2 = c'.$$

Now, if one of a' or b' is a QR (WLOG a') then $a'x^2$ is free to be any QR, which means we can set $a'x^2 = d^2$ and $y = 0$.

The other case is where a' and b' are not QRs. In this case we look back at the original equation, where a and b are QRs and c isn't. Then, ax^2 and by^2 are arbitrary QRs. By Fermat's two squares theorem, $u^2 + v^2$ can be any prime number that is $1 \pmod 4$, and by Dirichlet's theorem, there are infinitely many such primes in every nonzero residue class modulo p. Pick a prime q that is congruent to both $c \pmod p$ and $1 \pmod 4$ (which exists either by CRT or, if $p = 2$, trivially, since we already dealt with $c = 0$), apply the theorem for $ax^2 = u^2$ and $by^2 = v^2$ (equality in $\mathbb{F}_p$) and win. □

Problem 1.8

Show that there is a positive term of the Fibonacci sequence that is divisible by 1000.

Solution. We prove the well-known fact that the Fibonacci sequence is periodic. First, by Pigeonhole on the pairs (F_i, F_{i+1}) modulo 1000, some pair has to appear twice, i.e. $(F_n, F_{n+1}) \equiv (F_m, F_{m+1})$ modulo 1000 for some $m > n$. Then, we can backfill the numbers before F_n to find that the sequence is purely periodic (rather than just eventually), which means some nonzero F_n that is congruent to $F_0 = 0$ mod 1000. □

Problem 1.9

Let m be a positive integer. Prove that among any $2m + 1$ distinct integers of absolute value less than $2m - 1$, there exist three whose sum is zero.

Solution. If one of the integers is zero, we're done because one of the pairs $(1, -1)$, $(2, -2)$, $\ldots$, $(2m - 2, -2m + 2)$ must also be selected.

Now, suppose none of them are zero. Then, say there are k positive integers in the set, $a_1 < a_2 < \cdots < a_k$, and clearly $k \geq 3$. Consider the sums

$$a_1 + a_2 < a_1 + a_3 < a_2 + a_3 < a_2 + a_4 < a_3 + a_4 < \cdots .$$

There are $2k - 3$ of these. We want to show that one of these sums is equal in absolute value to one of the chosen negative numbers. Since there are $2m - 2$ choosable negative numbers and $2m+1-k$ chosen negative numbers, $k-3$ negative numbers are not chosen. However, $2k-3 > k-3$, meaning one of the sums must be equal in absolute value to one of the chosen negative numbers, as desired. □

Remark. Absolute value *at most* $2m-1$ also works.

Problem 1.10

Let $P_1, \ldots, P_{2m}$ be a permutation of the vertices of a regular polygon. Prove that the closed polygonal line $P_1P_2 \ldots P_{2m}$ contains a pair of parallel segments.

Solution. Let $\omega = \exp(\frac{\pi i}{m})$, so that the vertices of the $2m$-gon are powers of ω. Then, if $P_j = \omega^{x_j}$ for $1 \le x_j \le 2m$ and for all j, consider the permutation $(x_1, \ldots, x_{2m})$ of $(1, 2, \ldots, 2x)$. The main idea is the following fact about complex numbers on the unit circle:

Claim 1.10.1 — If a, b, c, and d lie on the unit circle centered at 0 in the complex plane, then the segment joining a and b is parallel to the segment joining c and d if and only if $ab = cd$.

Proof. The segments being parallel means that the expression

$$\frac{b-a}{d-c}$$

is real, since the numerator can be thought of as a vector along the line through a, b and the denominator as a vector along the line through c, d. This means the expression is equal to its own conjugate ($z = \overline{z}$ if and only if $z \in \mathbb{R}$), so

$$\frac{b-a}{d-c} = \overline{\left(\frac{b-a}{d-c}\right)} = \frac{\overline{b}-\overline{a}}{\overline{d}-\overline{c}} = \frac{\frac{1}{b}-\frac{1}{a}}{\frac{1}{d}-\frac{1}{c}}$$

because $|z| = 1$ means $z\overline{z} = 1$, i.e. $\overline{z} = \frac{1}{z}$. This simplifies to $ab = cd$ as desired. □

Since $\omega^{x_a} \cdot \omega^{x^b} = \omega^{x_a + x_b}$ where the exponent is only relevant as far as its residue modulo $2m$, we just need to show that there exist u and v such that

$$x_u + x_{u+1} \equiv x_v + x_{v+1} \pmod{2m},$$

noting that x_u, x_{u+1}, x_v, and x_{v+1} must all be distinct because clearly segments sharing a vertex can't be parallel, as each of the P_js are distinct. Now, if each $x_k + x_{k+1}$ was distinct mod $2m$ so that

$$\{x_k + x_{k+1} : 1 \le k \le 2m\} = \{1, 2, \ldots, 2m\},$$

then the sum of the $x_k + x_{k+1}$ is on one hand

$$2 \cdot (1 + 2 + \cdots + 2m) = 2m(2m+1) \equiv 0 \quad (\text{mod } 2)m$$

because each element of the permutation appears twice, but on the other hand is equal to

$$1 + 2 + \cdots + 2m = m(2m+1) \equiv m \pmod{2m}$$

by the distinctness assumption, but this is a contradiction. Thus two of the $x_k + x_{k+1}$ are identical, which is what we wanted. □

Problem 1.11

The points of the plane are colored with one of a finite number of colors. Prove that some rectangle has all vertices the same color.

Solution. We just focus on $\mathbb{Z}^2$. Suppose there are n colors, and consider the $n^{n+1}+1$ sets of points

$$S_k = \{(i,k) : 1 \le i \le n+1\}$$

where $1 \le k \le n^{n+1}+1$. Then, by pigeonhole, some two S_ks have identical color patterns; that is, there exist a, b such that (i,a) and (i,b) have the same color for all $1 \le i \le n+1$, because there are only n^{n+1} possible colorings for any S_k. To finish, we see that within S_a, there exist two points of the same color because there are $n+1 > n$ points in S_a: call these (c,a) and (d,a), and then we are done because (c,a), (d,a), (c,b), (d,b) make a monochromatic rectangle. □

Problem 1.12

Inside the unit square lie several circles, the sum of whose circumferences is at least 10. Prove that infinitely many lines intersect at least four circles.

Solution. Let the unit square have vertices $(0,0)$, $(1,0)$, $(0,1)$, $(1,1)$. Then, for each $0 < r < 1$, let $f(r)$ be the number of circles which have an interior point with x-coordinate r. The integral of f over $(0,1)$ is then $\frac{10}{\pi} > 3$, so there exists a set of r-values with nonzero measure for which $f(r) = 4$, the end. □

2 Invariants

Problem 2.1

TODO fill in

Problem 2.2

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